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EV-Insights: open source framework for electric vehicle charging data processing, analysis, and forecasting

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Abstract
The rapid growth of electric vehicle adoption presents new challenges for distribution system operators and charging station owners. Distribution system operators must correctly manage charging demand, optimize infrastructure usage, and support grid stability, while charging station owners seek to analyze user behavior and predict future demand to improve operational efficiency. Meeting these goals requires accurate analysis and forecasting of charging behaviors. A major bottleneck, however, lies in the heterogeneity of available electric vehicle charging datasets: each dataset comes with its own structure, quality issues, and missing information, requiring time-consuming and error-prone preprocessing before any analysis or forecasting can be performed. To overcome this limitation, we introduce EV-Insights, an open-source framework designed to provide standardized services for data ingestion, preprocessing, analysis, and forecasting by supporting the integration of real-time data, synthetic data, and public datasets. Once data is integrated, users can easily generate insights on charging behavior and extend the framework with new analyses or forecasting models through modular interfaces. We evaluated EV-Insights using seven real-world public datasets comprising over 3 million charging sessions, demonstrating its potential to uncover valuable insights and support informed decision-making. Ev-Insights is available as open source at https://github.com/EV-Insights

Keywords Electric vehicles (EVs), Charging data, Open source framework, Data processing, Data analysis, Forecasting

Introduction

The widespread adoption of electric vehicles (EVs) is redesigning the way we think about transportation and energy consumption. This transformation presents significant challenges for both distribution system operators (DSOs) and charging station owners (CSOs). DSOs must address challenges arising from uncontrolled or poorly coordinated electric vehicle charging, such as congestion in the distribution network, local voltage regulation problems [1–3], and the introduction of harmonics [4, 5]. At the same time, CSOs are increasingly focused on optimizing infrastructure usage, reducing operational

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costs, and enhancing service offerings through the analysis of user behavior and demand forecasting [6].

Several studies have explored charging behaviors and usage dynamics to better understand how, when, and where EVs interact with the grid-for instance by identifying spatial and temporal usage patterns, classifying user profiles, and examining how factors such as location, time of day, or pricing schemes influence charging behavior. [7] applied clustering and statistical modeling techniques to segment EV drivers based on their charging sessions, that eliminates the need to recluster each time a new user joins the charging network. More recently, [8] analyzed charging session data-covering approximately 7000 stations in Canada-to characterize EV users' charging behaviors, finding charging patterns variations across different charging point type (i.e. public and residential). In a complementary approach, [9] focused on the usage of charging stations themselves-covering in this case about 3000 charging stations in Switzerland-examining patterns of utilization and user interactions to support the planning, deployment, maintenance, and potential expansion of charging infrastructure. Several studies have also focused on developing advanced forecasting approaches to predict charging demand and support operational planning. [10] proposed a deep spatio-temporal framework based on graph convolutional networks to model spatial correlations among charging stations, improving short- and long-term demand predictions. [11] explored the combined forecasting of photovoltaic generation and EV charging demand to enhance microgrid management, showing how accurate predictions can significantly affect energy scheduling and system efficiency. Beyond these examples, a growing number of works have addressed these topics. Comprehensive overviews are provided by [12–14]. However, the impact of these advanced forecasting approaches is often limited by the scarcity and fragmentation of available EV charging datasets.

Addressing these challenges is crucial to ensure a reliable and efficient energy system. The complexities of integrating EVs into existing infrastructure highlight the need for innovative solutions and strategic planning, requiring access to a vast amount of high-quality data, which is often not readily available or operable. Indeed, data may be limited in size, both in terms of the number of users and charging stations they cover, as well as in the number of samples collected over time. This scarcity of extensive datasets poses a challenge for researchers and industry stakeholders aiming to develop robust analytical models and forecasting tools [15]. To address this limitation, one approach is the generation of synthetic data. Tools such as "emobpy" allow the creation of realistic EV mobility and charging time series based on limited empirical data, simulating vehicle availability and electricity demand for large fleets [16]. Similarly, comprehensive synthetic datasets have been developed to represent residential electricity prosumers, including EV charging, photovoltaic generation, and energy storage systems, using a combination of empirical data and generative adversarial networks (GANs [17]) to produce synthetic time series [18]. Another example is FlexiGen, an open-source stochastic dataset generator that supports both vehicle-to-grid (V2G) and unidirectional (V1G) demand response scenarios, encompassing realistic electric vehicle usage, user behaviors, and flexibility scenarios across both residential and workplace charging routines [19].

Another potential solution to mitigate this limitation is leveraging publicly available datasets, often provided by research institutions, governmental agencies, or industry collaborations, that can serve as valuable resources for conducting analyses and

validating models. However, one limitation is the lack of standardization in data collection, which reduces their usability. Each dataset differs in structure, semantics, and completeness, and often requires extensive manual preprocessing-such as cleaning, normalization, and mapping to a common format-before it can be used. This process must be repeated from scratch for every new dataset, requiring extra effort, making datasets harder to compare, and reducing reproducibility. This makes integration challenging and requires careful consideration of data preprocessing techniques to ensure comparability across different sources. Poorly handled preprocessing can significantly degrade the performance of predictive models and lead to erroneous conclusions [20]. These inaccuracies, in turn, can result in suboptimal infrastructure planning, inefficient energy management, and misguided investment decisions. Addressing these challenges requires continuous effort from those analyzing these data to implement advanced preprocessing and analysis techniques, ensuring the extraction of meaningful insights from raw data.

After introducing the motivation and objectives of this work (Section 2), the remainder of the paper first describes the EV-Insights framework (Section 3), including its system architecture (Subsection 3.1) data model (Subsection 3.2), and main services (Subsection 3.3). In addition, the paper highlights how the functionalities of the core modules (data ingestion, analysis, and forecasting) support different types of users, by illustrating common use cases (Section 4). Finally, it concludes with a summary of the main findings and a discussion of potential directions for future development (Section 5).

Contribution and related work

The core objective of EV-Insights is to provide a set of essential services to address the challenges outlined. Specifically, EV-Insights offers functionalities for data ingestion, preprocessing, analysis, and forecasting. By leveraging these capabilities, stakeholders can gain useful insights into EV charging patterns, optimize infrastructure usage, and enhance grid stability through more precise demand prediction.

Previous contributions mostly focus on single services for single datasets or are strictly related to single pilot projects. [21] focuses on a large dataset with 72k+ sessions from more than 2000 EV users. Another large open source dataset is described in [22], which provides charging sessions collected from two workplace charging sites in California. Pilot projects releasing open charging datasets are described in [23–29]. [30] describes power consumption prediction using data coming from public charging outlets in Colorado and from public charging terminals in Kerala, India. To our knowledge, these datasets are not made publicly available. Interestingly, to overcome data scarcity, [31] proposed ACN-Sim, a data-driven, open-source simulator that simulates real-world charging systems. On a similar note, [32] describes methods obtain accurate synthetic charge profiles.

More recently, thanks to the development of Large Language Models (LLMs), new perspectives on LLM-based analysis and prediction tools have emerged [33, 34]. Along these lines, [35] proposed ChatEV, an LLM-based EV charging demand predictor. Furthermore, LLM-based approaches have gained traction in the field of time series analysis. A central challenge in applying LLMs, originally trained on textual data, is bridging the modality gap. [36] survey techniques for harnessing LLMs across diverse domains and multimodal time series datasets. Similarly, [37] offer a systematic overview, outlining a general pipeline and classifying LLM-based methods. Finally, [38] present a

position paper arguing that LLMs have the potential to revolutionize time series analysis by enabling universal analytical intelligence, as LLMs can act as data/model enhancers, superior predictors, and next-generation agents. While promising and full of potential, this line of research remains in its early stages. The use of generative AI for forecasting and analytical tasks still faces significant challenges, particularly in terms of efficiency, computational cost, and effective prompting-especially when dealing with long sequences of high-precision numerical data or complex multivariate time series. Employing LLMs as enhancers or integrated tools can introduce substantial time and cost overheads, which is especially problematic for tasks requiring real-time responses or involving large-scale signal processing. Overall, LLM-centric approaches to time series analysis may result in high environmental and computational costs, despite the availability of well-established, cost-effective, and accurate traditional methods.

Unlike existing tools, which are typically developed for single pilot projects and require substantial effort to adapt to new data, EV-Insights provides a generalized preprocessing and ingestion layer. Once a simple translation function is defined, any dataset-public, private, or synthetic-can be easily integrated into the system. Similarly, thanks to its modular design, new analysis or forecasting methods can be added with minimal effort, without the need to rebuild the entire pipeline. This enables researchers to focus on designing and comparing algorithms rather than spending significant time on data cleaning and preparation. The ability to process simultaneously multiple datasets allows a more comprehensive understanding of EV charging behavior across different regions, user groups, and infrastructure types. In contrast to previous works that focus on isolated aspects of EV charging session data analysis-such as demand forecasting, user classification, or grid impact assessment-EV-Insights provides a unified framework that supports all these tasks holistically.

Standardizing data access and preprocessing, EV-Insights lowers the entry barrier for developing new algorithms, making it easier for researchers and practitioners to experiment and design novel models without building a data pipeline from scratch.

At the time of writing, EV-Insights comprises four main service modules: Data Administration, Data Ingestion, Analysis, and Forecasting. Due to its fully modular design, additional modules can be integrated to extend its functionality. Data Administration enables users to manage and organize information from various datasets and charging sessions, ensuring structured and efficient data handling. Data Ingestion is equipped to read public datasets from multiple pilot projects, facilitating the integration of real-world charging session data from diverse EVs. Extending the existing dataset ingestion to new pilots involves a minor effort into data translation from existing sources (e.g. CSV, XLSX and JSON files that are managed natively). Analysis module is capable to visualize data and compare different datasets. It enables users to gain insights into charging patterns, user behavior, and charging station events. As for Data Ingestion, adding custom analysis views is made easy by a standardized interface. Finally, Forecasting module currently offers four forecasting capabilities: for individual user, it exploits tree boosting methods to predict next day charging session duration and charging session energy demand while for charging stations, tree boosting methods are used to predict next day the total charging demand and the number of required charges. Similar to the previous, the Forecasting module also supports the easy integration of custom forecasting modules through the standardized interface.

EV-insights framework design

EV-Insights is a framework aimed at optimizing the management of EV charging data. This section offers a detailed overview of the system architecture, along with a description of the data model and the main services offered: Data Administration, Data Ingestion, Analysis, and Forecasting.

System architecture

The system architecture of the EV-Insights (Figure 1) is designed to create a robust, scalable, and efficient environment for managing EV charging data. Employing a microservices approach, each component is designed to fulfil a specific function. This structure promotes modularity, maintainability and allows for independent scaling of each component as needed. The system exposes its functionalities via a RESTful Application Programming Interface (API) and a Command-Line Interface (CLI) enabling both automated and manual interactions. APIs are built using FastAPI [39] and Uvicorn [40], ensuring high performance and responsiveness. This interface allows external applications or users to interact with EV-Insights. The complete list of available APIs at the time of writing is provided in Appendix A. The CLI interface lets users to interact directly with the services, bypassing the web interface when needed (e.g., for batch jobs or cron-like scheduling). The core of EV-Insights is composed of four main services, each responsible for specific functionalities and designed to be stateless, communicating with other modules to retrieve and persist data as needed:

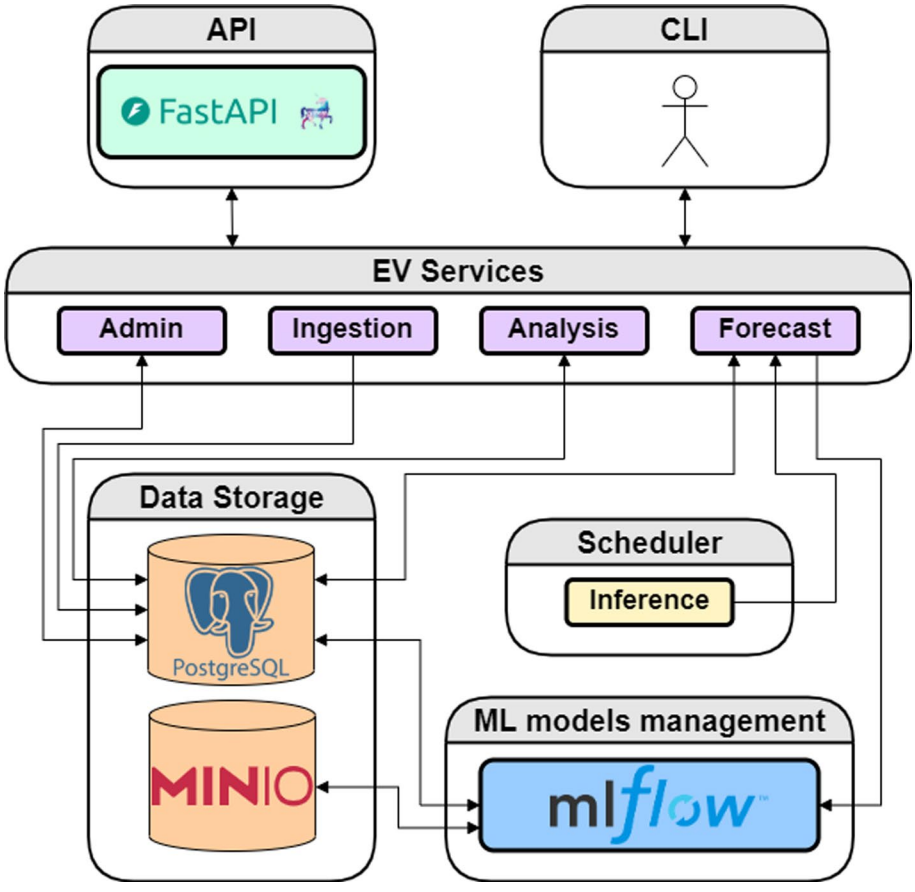

Fig. 1 Ev-Insights System Architecture

- Admin: handles database initialization, and administrative tasks such as cleaning or resetting the database.
- Ingestion: manages the data ingestion pipeline, including reading raw data from various sources, preprocessing, validation, and storage.
- Analysis: performs exploratory and statistical analysis of the charging session data, helping to understand behavioral patterns and trends.
- Forecast: implements predictive models to estimate future charging behavior, such as connection times and energy demand.

Two back-end storage systems are used: PostgreSQL [41] and MinIO [42]. PostgreSQL is used as the main relational database, storing structured data such as metadata (dataset, charging stations, users and ev details), charging sessions and predictions. It provides a robust and reliable solution, offering full SQL compliance, strong data integrity, and efficient support for complex queries. Thanks to the modular architecture of the EV-Insights framework, where the database interface is abstracted from the rest of the system, the storage system can be easily replaced by implementing a custom database connector. MinIO is used for object storage, handling predictive models, model artifacts, and experiment logs. MinIO offers S3 compatibility, making it suitable for scalable and cloud-native deployments. EV-Insights integrates MLflow [43] to manage the entire lifecycle of machine learning models which is responsible to handle models, model training metadata, experiment tracking, versioning, and storage of serialized model files. This integration enables structured model comparison, supports ex-post performance analysis, and facilitates iterative refinement, while ensuring reproducibility of forecasting experiments across heterogeneous datasets. Finally, the Scheduler automates inference tasks. It periodically runs prediction tasks by retrieving the latest model version from MLflow, running the model on newly ingested data, storing prediction results back into the PostgreSQL database. This decouples real-time interaction (via API or CLI) from background inference processes, ensuring predictions are continuously updated.

Data model

The data model is defined through a comprehensive SQL script, designed to create and structure the necessary database schema. The script begins by establishing the new schema, ensuring that all subsequent tables and relationships are organized within this dedicated namespace. The script then focuses on defining the core tables that will store essential data related to EV charging sessions, charging stations, datasets, users, and forecasting information. Each table is structured with specific fields and constraints to maintain data integrity and facilitate efficient data retrieval. The Entity-Relationship (E/R) model shown in Figure 2

illustrates the structure and relationships of the tables and data entities within the PostgreSQL database, which serves as the backbone for storing and managing EV charging station data. A brief overview of each table is provided below, while a comprehensive field-by-field description is available in Appendix B.

- ChargingSession: this table records individual charging sessions, capturing vital information such as the state of charge upon plugging in (plug_in_soc), timestamps for plug-in and plug-out events, and the amount of energy supplied during the

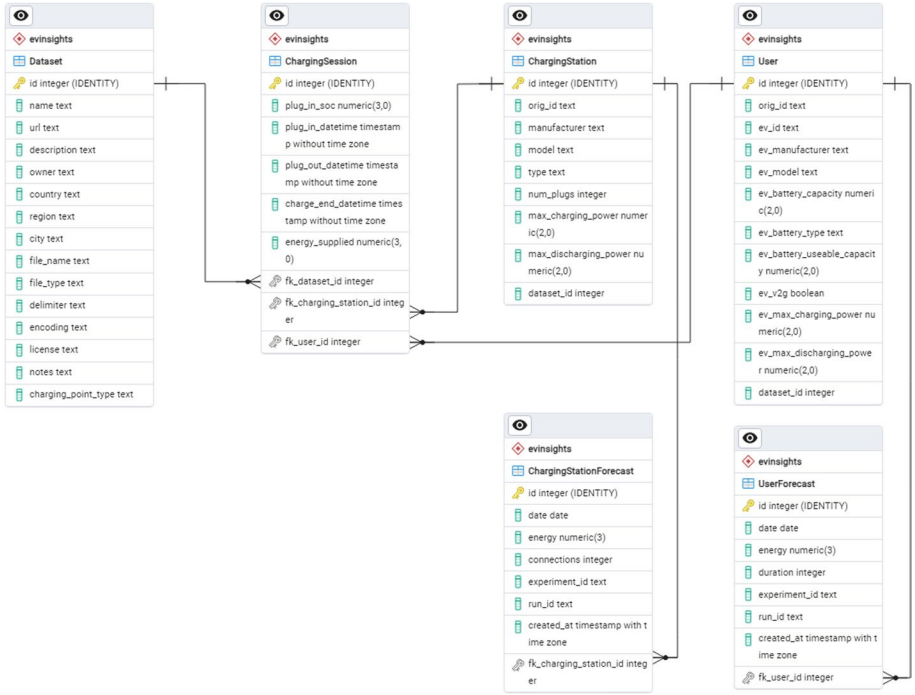

Fig. 2 Ev-Insights Data model

session. The table also includes foreign key references to link each session to the corresponding charging station, dataset, and user.

- ChargingStation: this table contains details about each charging station, including the manufacturer, model, number of plugs, and maximum charging and discharging power capabilities. It serves as a reference point for the charging sessions recorded in the ChargingSession table.
- Dataset: this table provides metadata for the datasets used within the project, including descriptive fields such as name, URL, owner, and geographic details. This ensures that all data sources are well-documented and easily identifiable.
- User: this table captures information about users, including their electric vehicle details, such as manufacturer, model, battery capacity, and charging power. It establishes a link between users and their charging sessions, facilitating user-specific data analysis and forecasting.
- ChargingStationForecast: this table is designed to hold forecasted data related to charging stations, detailing expected energy usage and number of connections.
- UserForecast: this table stores forecast data specifically for users, detailing expected energy usage and charging duration.

Throughout the script, foreign key constraints are established to maintain the relationships between tables, ensuring referential integrity. The script also includes error handling measures by dropping existing tables before creating new ones, thus preventing conflicts and maintaining a clean database structure.

Services

This section presents a comprehensive overview of the various functionalities integrated into EV-Insights. These services are designed to facilitate the efficient management, analysis, and forecasting of EV charging session data.

Admin

The Admin module serves as a tool for managing the database content, providing functionalities for both initialisation and maintenance. Initialisation involves setting up the whole structure required for storing and managing data using a dedicated SQL script, which creates the necessary tables, defines the appropriate fields, and establishes relationships between data entities. In addition to the initialization function, the Admin module also includes a service for deleting specific datasets from the database. This feature allows administrators to maintain a clean and organised database by removing unnecessary or outdated data entries. Although these functions are incredibly useful, especially during the development phase, when rapid iterations and adjustments are common, they involve sensitive operations that could significantly impact the integrity of the database. To address this concern, the implementation of additional authorization controls are planned for future releases.

Ingestion

The Ingestion module is responsible for managing the complete data acquisition process. This encompasses reading, pre-processing, cleaning, and inserting data into the PostgreSQL database. It is designed to handle a variety of data sources, enabling the flexibility required to accommodate both entire datasets and real-time, point-specific data as it becomes available. The preprocessing pipeline plays a central role in the framework by enabling the automated and reproducible integration of heterogeneous datasets into the proposed unified data schema. The data model has been designed to capture details that are often missing in currently available public datasets, such as EV static information (e.g., battery capacity, maximum charging and discharging power) and dynamic data, including the battery state of charge at the time the EV is plugged in. Given the heterogeneity of available datasets, which is expected to persist due to legacy installations and custom data collection practices, this preprocessing module provides a practical solution to map diverse datasets into the common schema. This approach significantly reduces the manual effort typically required for data preparation and ensures that the resulting data is consistent, comparable, and reusable across different analyses and studies. For public datasets, the ingestion process begins with a pre-processing phase. The module is able of reading files in multiple formats, including CSV, XLSX, and JSON, ensuring compatibility with various data sources. Each dataset undergoes a specific preprocessing routine used to standardise the data according to the proposed data model. This standardisation is essential for maintaining consistency and reliability in the database. During this phase, data cleaning is also performed, where entries with null values or anomalies are removed to ensure that only high-quality data is retained for analysis. In the case of real-time data, it is assumed that the incoming information adheres to a pre-agreed format. This standardization facilitates seamless integration into the system, allowing for immediate processing and analysis without the need for extensive preprocessing steps. Overall, the Ingestion module guarantees that data is accurately and

efficiently prepared for subsequent analysis and forecasting tasks. Several public datasets have already been imported into the system, with dedicated ingestion functions implemented and tested to automate their processing. These datasets cover diverse geographic regions, time periods, and charging point types. Table 1 summarises the key characteristics of these datasets.

Analysis

The Analysis module provides a framework for managing and executing different types of data analyses in a standardized way. It defines a base class that ensures consistency across various analyses and includes utility functions to initialize and dynamically load specific data and functions based on configuration files formatted in JavaScript Object Notation (JSON) [44]. Each analysis follows a consistent structure, starting with data retrieval, validation and preparation. Custom settings can be applied to run the analysis with specific requirements, such as adding new features or filtering data. The main logic of the analysis is implemented in the "run" method, which processes the data and generates outputs like visualizations or summary statistics. Results can be saved in various formats, including images, JSON, XLSX, and PDF, depending on the analysis type. Each analysis can be executed on individual datasets or across all datasets imported into the database. Thanks to the standardized data format and to a configurable parameter, datasets can be merged, enabling more insights from combined data sources. In most analysis, an optional clipping range can be applied to make some filters during data retrieval (i.e. by energy demand, plug-in datetime, etc.). The following list describes the available analysis modules currently implemented in EV-Insights that can be executed using the framework. Examples of visual outputs for a selected subset are also included to illustrate the kind of insights they provide.

- Animated energy duration: creates an animated visualization that shows the relationship between energy supplied and plug duration which dynamically shows how evolves over time.
- Number of charges by hour: shows the distribution of charging events across different hours of the day by counting the number of occurrences for each hour of the day. Figure 3 shows the hourly distribution of plug-in and plug-out events in the OLEV dataset. A clear daily pattern emerges, with most EVs plugging-in during the evening and unplugging in the morning, indicating that the majority of charging occurs overnight. These temporal trends are crucial for identifying peak demand periods and can inform planning strategies for energy allocation and grid balancing.
- Number of charges by weekday: shows the distribution of charging events across the days of the week by counting the number of occurrences for each day of the week.

Table 1 Public datasets supported by the ingestion module

Name	Country	Period	Charging Point Type	Samples	Ref.
ACN_Caltech	USA	2018–2021	Private (workplace)	31,424	[22]
AMB_Barcelona	ES	2019	Public	36,413	[32]
Elaad	NL	2019	Public	10,000	[24]
OLEV	UK	2017	Private (domestic)	3,173,236	[25]
BeLib	FR	2017	Public	3,351	[23]
Harvard_Dataverse	USA	2015	Private (workplace)	3,395	[29]
Norway_12loc	NO	2018–2021	Private (domestic)	35,000	[26]

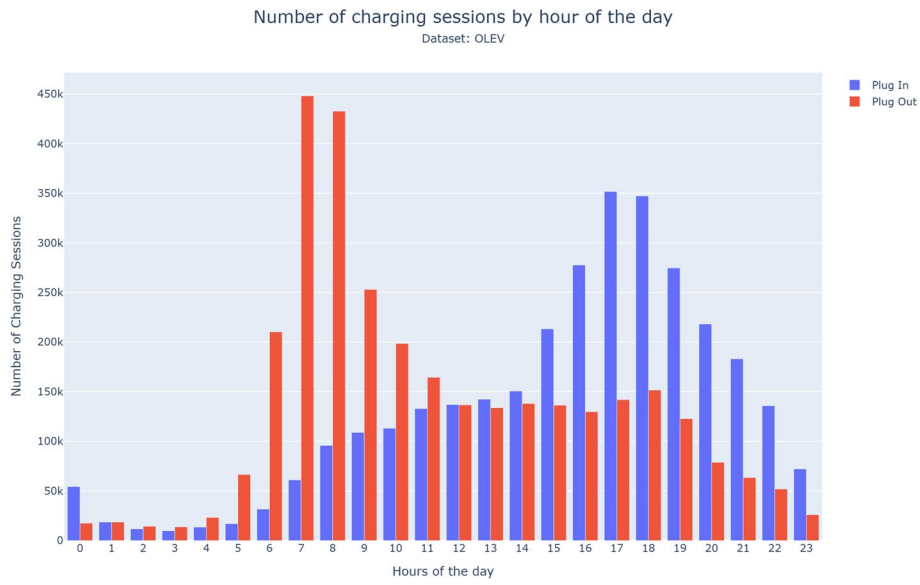

Fig. 3 Number of charging sessions by hour of the day in the OLEV dataset

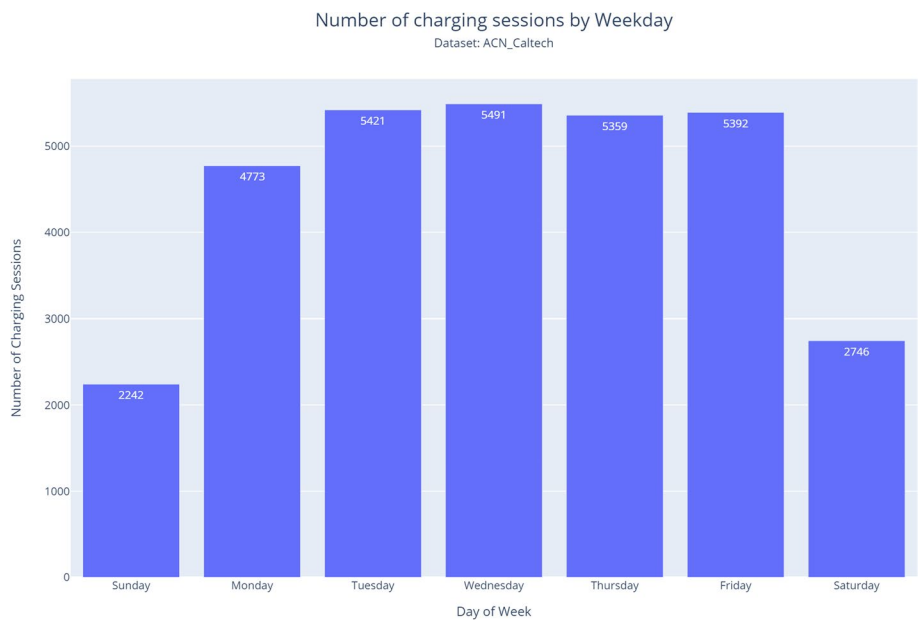

Fig. 4 Number of charging sessions by weekday in the ACN_Caltech dataset

Figure 4 illustrates an example based on the ACN Caltech dataset. The data reveals a higher concentration of charging sessions on weekdays, while weekends show a significant drop. This pattern is consistent with usage in institutional or workplace environments, where electric vehicles are primarily charged on working days. This is consistent with the nature of the ACN dataset, which collects data in a parking garage at the Caltech campus.

- Distribution of energy demand: creates an histogram to show how energy demand is distributed. The histogram in Figure 5 reveals that the majority of charging sessions involve relatively low energy supply, with a peak between 5 to 10 kWh. As the supplied energy increases, the frequency of occurrences decreases, forming a long-

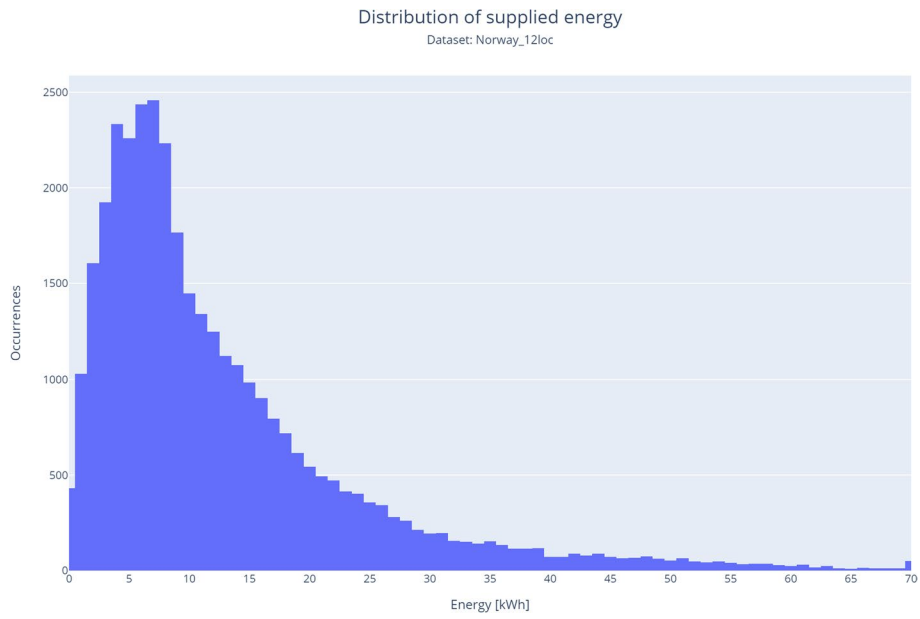

Fig. 5 Distribution of supplied energy in the Norway_12loc dataset

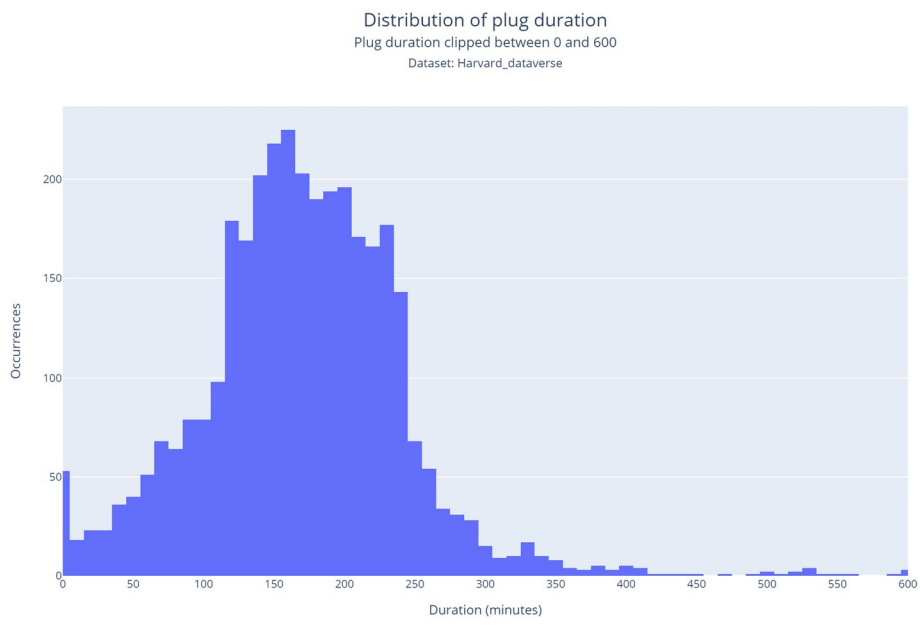

Fig. 6 Plug duration distribution in the Harvard_Dataverse dataset

tail distribution. This insight shows typical charging behaviors for electric vehicles in the Norway_12loc dataset.

- Energy supplied by plugin hour: examines the relationship between the energy supplied and the hour of the day when the plug-in events occur by showing the results through box plots.
- Total energy supplied by weekday: examines the total energy supplied across the days of the week.
- Plug duration: examines the distribution of plug durations and shows an histogram to visualize the frequency of these durations. Figure 6 represents the distribution of

plug durations from the Harvard_Dataverse dataset. Most sessions are shorter than 4 hours, with a clear peak during the first two hours. The distribution declines rapidly for longer durations, indicating that short or partial charging sessions are more common.

- Plug duration by plug-in hour: examines the relationship between plug durations and the hour of the day to illustrate how plug durations vary across different hours.

This list of analyses is not intended to be exhaustive. It serves as a starting point and will continue to evolve over time expanding and adapting to meet new requirements and user needs.

Forecast

The Forecast module provides a structured framework for building and using predictive models within the EV-Insights platform. It provides functionality to create and use forecasting models for predicting specific outcomes, such as energy consumption or charging duration. Similar to the Analysis module, it is implemented to ensure consistency across different forecasting modules, by starting with data validation and preparation and the possibility to apply custom settings to run the module with specific requirements, such as adding specific features or filtering data. Each implementation must include the two main function: train, for generating new models or updating existing ones, and predict, for producing inference results.

A predefined set of forecasting implementations is provided, with the included algorithms intentionally kept simple and text-book like, as the aim of this work is not to identify the optimal predictive model for these forecasting tasks, but to provide a framework that facilitates easy creation and integration of prediction tools.

- LightGBM Station Charges: predicts the number of charges of a specific charging station for the next day. Predictions are provided by using the LightGBM [45] machine learning algorithm. Table 2 reports results related to a selection of 6 charging stations with more than three thousands recorded charging sessions. The experiments were conducted on the "AMB_Barcelona" dataset, covering a period of 1 year, using a split where 90% of the data was allocated for training and 10% for testing. MAE and Accuracy metrics for the Daily Demand (DD) and the Number of Charging Sessions (CS) have been reported. We can observe that the number of daily charging sessions is fairly predictable with errors averaging way below four charging events. The total daily demand is less predictable with the current model and the available data.
- LightGBM Station Energy: predicts the energy demand of a specific charging station for the next day. Predictions are provided by using the LightGBM machine learning algorithm.

Table 2 Charging Station – Daily Demand (DD) and Charging Sessions (CS)

Id	Charging sessions [#]	DD MAE [kWh]	DD Accuracy [%]	CS MAE[#]	CS Accuracy [%]
1	5139	51.3	42.4	3.0	70.4
2	4686	40.2	77.4	3.2	79.1
3	4377	46.0	59.4	2.9	71.6
4	3824	39.4	72.0	2.9	78.0
5	3706	42.8	28.0	2.3	53.0
6	3659	30.8	61.2	2.2	73.7

- XGBoost Charge Duration: predicts the charge duration of the next charging session of a specific user. Predictions are provided by using the XGBoost [46] machine learning algorithm.
- XGBoost Energy: predicts the energy demand of the next charging session of a specific user. Predictions are provided by using the XGBoost machine learning algorithm. In this case, the investigation has been restricted to users with a sufficient history of charging sessions in the datasets. The experiments were conducted on the "Harvard_Dataverse" dataset, covering a period of 1 year, using a split where 90% of the data was allocated for training and 10% for testing. To evaluate training performance we report the MAE in minutes and Accuracy as a percentage. Table 3 reports training results for individual user forecasting related to Energy Demand (ED) and Charging Duration (CD). As seen in the results, individual user forecasting is extremely challenging. In particular, forecasting the connection duration, i.e. the amount of time the EV will remain connected for charging, proved to be very hard with average errors ranging around one hour and very low accuracies in some cases. Instead, the energy demand is a bit more predictable (except one case) with errors ranging within 1 kWh which makes the predictions usable for decision making.

This list of forecasts is not intended to be exhaustive. It serves as a starting point and will continue to evolve over time expanding and adapting to meet new requirements and user needs.

Use cases

This section outlines the main use cases that may arise when using EV-Insights, which is designed to support different types of users and facilitate its adoption and effective use. First, EV-Insights allows single users such as DSOs, charging station operators, or private individuals to upload their datasets, either in bulk (if a dataset is already available) or as individual tables (if they need to set up a real-time, continuous data import), and ingest them into a centralized database, which could be hosted locally or remotely, depending on user preferences. In addition, single users can perform detailed analyses of the data ingested, generating statistics such as energy usage, charging session durations, and usage patterns, visualized through customizable reports and graphs. Researchers can leverage the framework to work with multiple datasets simultaneously, conducting cross-dataset analyses and predictions. Finally, developers can expand the framework by introducing new services to address specific needs. A dedicated tutorial for each use case can be found in the "doc/tutorials" folder of the EV-Insights GitHub repository [47].

Table 3 Individual user – Energy Demand (ED) and Charging Duration (CD)

Id	Charging sessions [#]	ED MAE [kWh]	ED Accuracy [%]	CD MAE[min]	CD Accuracy [%]
1	192	1.2	63.1	58.6	19.0
2	170	2.6	38.1	56.6	4.2
3	160	1.3	51.6	31.2	67.3
4	147	1.3	77.3	80.9	59.1
5	141	1.1	88.5	38.2	69.7
6	130	1.2	74.4	71.5	64.8
7	120	0.9	83.9	61.6	56.7

Ingesting data into a database

This section describes the steps a user must follow to ingest charging data into a database.

Bulk ingestion

The Bulk Ingestion process is designed to handle and process one or multiple datasets with different file formats and content in one go. This is mainly used to ingest an entire dataset, such as a public dataset. For this to work effectively, the user must ensure that the dataset file are properly stored in the appropriate input directory (e.g., data/input/public_datasets/<dataset_name>) and for each dataset a set of basic function aimed to prepare the data in the data model format is prepared (some examples are available in the src/interfaces/file_interface.py file). Additionally, the data/input/datasets_details.csv file must be fully updated. This file contains important metadata about all the datasets being ingested. It includes fields like the dataset name, description, file name, type (e.g., CSV, JSON, or Excel), delimiter, encoding, and other details such as the owner, region, license, and any relevant notes, essential as they provide the system with the information it needs to locate, open, and correctly read each file. For a step-by-step guide, refer to the dedicated tutorial in the GitHub repository, available in the "Ingestion bulk.md" file.

Table ingestion

The Table Ingestion process allows adding new pieces of information to specific tables in the system, such as datasets, users, charging stations, and charging sessions. This process is designed to be flexible and efficient so that new data can be added without reprocessing everything in the database allowing continuous data ingestion. To begin, each type of data to be inserted (e.g., new datasets, users, charging stations, or charging sessions) must be stored in its own file. These files should be prepared following the required structure for each category and placed in the appropriate input folder specific to the pilot or project (e.g., data/input/pilots/<pilot_name>). The system then processes the data through the ingestion service. This service works in an orderly manner, handling each category of data one at a time in the following order:

- Datasets: if a dataset file is found, the system reads the file and extracts the dataset's details. It then adds this information to the database, ensuring the new dataset is registered and ready for use.
- Users: if there is a file with new users, these are added to the database next. This ensures that all new users are accounted for and linked to other data if necessary.
- Charging Stations: the same applies to charging stations, if there is a file with new charging stations, the related records are saved so the database.
- Charging Sessions: finally, the system processes new charging session files, adding details about new charging events.

This process ensures each type of data is handled separately but in a logical order to maintain consistency across the database. The ingestion process can be performed either automatically, with a scheduler or automation that detects new files, or manually by running a script when it is necessary. For a step-by-step guide, refer to the dedicated tutorial in the GitHub repository, available in the "Ingestion table.md" file.

key	value
User id	10.0
Number of charges [#]	19
Plug duration total [min]	9045.83
Plug duration min [min]	282.48
Plug duration max [min]	550.68
Plug duration mean [min]	476.1
Energy supplied total [kWh]	371.00
Energy supplied min [kWh]	10.00
Energy supplied max [kWh]	30.00
Energy supplied mean [kWh]	19.53
Average number of days between charges [#]	5.5

Fig. 7 Extract of a user statistics provided in a PDF file

key	value
Dataset name	ACN_Caltech
Number of charges [#]	31424
Plug duration total [min]	10656737.07
Plug duration min [min]	2.07
Plug duration max [min]	14716.15
Plug duration mean [min]	339.13
Energy supplied total [kWh]	291194.00
Energy supplied min [kWh]	1.00
Energy supplied max [kWh]	78.00
Energy supplied mean [kWh]	9.27

Fig. 8 Extract of a dataset statistics provided in a PDF file

Single user

DSO, charging station owners, private users, or any interested party can leverage EV-Insights to process its dataset and extract valuable insights or predictions about charging behavior. The workflow is straightforward: after ingesting the dataset as described in the previous section, the user can run one or more pre-configured analysis by selecting the analysis type. In addition, if more specific information is needed, the user can adjust a few parameters, such as time or energy ranges or filter results by dataset, user, or station IDs. Once configured, EV-Insights dynamically loads the appropriate analysis module from its library, ensuring that the selected analysis aligns with the needs. It retrieves the relevant data from the database, validates it, and processes it according to the logic defined in the chosen module. The results, whether they are statistical summaries or predictions, are saved in a structured format and can be visualized for better interpretation in different file formats like images, spreadsheets or pdf documents.

An example of analysis that a user can perform using the implemented functionality is the generation of detailed statistics about their electric vehicle charging sessions. By providing the dataset reference, the user can configure EV-Insights to analyze their data. For instance, EV-Insights can calculate metrics like the total number of charging sessions, some basic statistics like total, minimum, maximum, average and standard deviation of plug durations and energy supplied. These metrics can also be computed for each dataset. Figures 7 and 8 show excerpts from a PDF report generated by EV-Insights,

illustrating simple statistics computed for an individual user and for an entire dataset, respectively.

Additionally, EV-Insights can provide insights into the user's charging behavior over different time periods. For example, it can calculate the same metrics for the past 1 day, 7 days, 30 days, or even a year. This allows the user to observe trends and patterns in their charging habits over time. The results of the analysis are saved in various formats, such as JSON, XLSX, PDF or PNG and can include visualizations like histograms or bar charts for easier interpretation. For a step-by-step guide, refer to the dedicated tutorial in the GitHub repository, available in the "Analysis stats service.md" file.

Researchers and data scientists

In this section, we examine another use case of EV-Insights, specifically designed for a distinct category of users: researchers and data scientists. These users typically require substantial amounts of data to conduct their research, which a single dataset often fails to provide. It is widely recognized that obtaining a single dataset containing a large number of charging session records is uncommon. Consequently, these users frequently engage in the process of identifying and merging multiple datasets to achieve a sufficient sample size. This phase of their work often demands significant resources due to the heterogeneity of the available data. EV-Insights aims to address this challenge by facilitating the efficient and resource-effective aggregation of diverse datasets. The steps required to add a new dataset include uploading the dataset to the designated input folder, adding its metadata to the dataset_details.csv file, and creating a small custom function for data import, which can be easily developed by referencing existing implemented functions, basically used for mapping the data with the database structure. The remaining infrastructure ensures that the data is read, processed and loaded into the database while maintaining consistency with other datasets. This allows researchers and data scientists to focus more on developing advanced analytical and forecasting methods that are directly relevant to their research objectives. For a step-by-step guide, refer to the dedicated tutorial in the GitHub repository, available in the "Analysis and forecasting services.md" file.

Developers

As an open-source framework, EV-Insights is also designed to engage developers who are interested in enhancing and extending its functionality. For instance, developers can implement custom functions to preprocess and transform datasets, ensuring seamless compatibility with the EV-Insights infrastructure. Additionally, they can create new customized input and output interfaces to support a broader range of file formats and databases and implement new APIs to provide additional services. Furthermore, they can integrate entirely new services, which can be incorporated into the existing framework with ease. To facilitate this, EV-Insights is developed following best practices in software engineering. These include principles such as modular design, which ensures that individual components can be developed, tested, and maintained independently; maintainability, achieved through clean, well-documented, and readable code; and high customizability, allowing users to adapt EV-Insights to their specific needs with minimal effort. For a step-by-step guide, refer to the dedicated tutorial in the GitHub repository, available in the "Developments.md" file.

Conclusions

In this work we presented EV-Insights [47], an open source framework designed to provide functionalities for data ingestion, administration, analysis, and forecasting by supporting the integration of EV charging data from different sources, including real-time data, synthetic data, and public datasets. Beyond offering a set of services, the main contribution of EV-Insights lies in its ability to standardize and automate data preprocessing across heterogeneous sources. Without such a framework, each dataset must be manually inspected, cleaned, and reformatted, a process that is not only time-consuming but also prone to inconsistencies and errors. By providing a unified ingestion and harmonization layer, EV-Insights significantly reduces this overhead, enabling datasets to be rapidly integrated and made ready for analysis and forecasting. A second key strength of the framework is its modularity and extensibility. The framework integrates a set of modules within a coherent structure, allowing for flexible use across research and operational contexts. EV-Insights aims to standardize workflows through consistent interfaces and configurations, providing a fast yet robust way to read and load data in a uniform manner. This enables reproducible experiments and easy extension. Its support for multiple data sources facilitates the rapid prototyping of new predictive algorithms and the implementation of new analysis, allowing users to explore and compare different solutions without the overhead of setting up a full data ingestion infrastructure from scratch. We support the use and extension of the library via hands-on tutorials shipped with the library on GitHub, we plan to continuously update these tutorials to support advanced functions and common use cases including data coverage, anomaly detection, data cleaning procedures, and the handling of specific data formats. We demonstrated the potential of EV-Insights through the integration of seven real-world public datasets, comprising over 3 million charging sessions. Although the current framework provides a rich set of functionalities, it could be further enhanced by incorporating additional factors that influence EV users' charging behavior, such as day-ahead electricity prices, local tariffs, or other contextual variables. Future developments will focus on expanding analytics and forecasting methods, both including the integration of external influencing factors and the introduction of a clustering service that can be designed to support forecasting algorithms in generating more accurate predictions. This service will help identify patterns such as groups of users with similar EV charging behaviors or sets of charging stations that deliver comparable amounts of energy within similar time windows, enhancing the accuracy and robustness of predictive models.

Appendix A: APIs

EV-Insights offers an initial suite of APIs designed to enable seamless access to the functionalities described. Ongoing platform development will allow additional APIs to be added. It is important to emphasize that functionalities not yet available through the APIs can still be fully accessed via the CLI.

- Admin:
  - init_db: initializes the database, setting up the necessary schema, tables, and constraints required for data storage and integrity. Any existing data will be deleted during the process.

  - delete_dataset: deletes a specific dataset from the database, a valuable function for maintaining database cleanliness and managing storage efficiently.

- Ingestion:
  - ingest_datasets: allows for batch insertion of data from one or more datasets into the database.
  - ingest_data: manages the continuous real-time data insertion process, making it possible to add new datasets, charging stations, users, or charging sessions. This feature is particularly useful for pilots needing to insert new data in real time, ensuring that the system has up-to-date information.

- Analysis:
  - stats: retrieves general statistics from the database, including metrics such as the number of entries in each table, data distribution summaries, and other key insights. This API provides a convenient way to access quick analytical insights on stored data.

- Forecast:
  - train: trains forecasting models on the available historical data, leveraging machine learning to generate predictive insights.
  - predict: offers two functionalities:

    schedule: generates predictions using the latest trained model version stored in MLflow and saves them to the database. This function is triggered by an internal scheduling service at configurable intervals (default: once per day) to ensure up-to-date predictions are regularly stored.
    query: queries the database to retrieve the most recent forecasted data saved by the scheduling service, enabling users to access the latest predictions on demand.

Appendix B: Data model technical specification

Table 4 Dataset table

Field	Description
id	The unique identifier of the dataset
name	The name of the dataset
url	The web address where the dataset can be accessed and downloaded
description	A brief summary of what the dataset contains
owner	The individual or organization responsible for the dataset
country	The country where the dataset was created
region	The specific region within the country that the dataset concerns
city	The city within the region that the dataset is associated with
file_name	The name of the file containing the dataset
file_type	The format of the file (e.g., CSV, XLSX, JSON)
delimiter	The character used to separate fields in the file
encoding	The character encoding used in the file (i.e. UTF-8)
license	The terms under which the dataset can be used or distributed
notes	Any additional information or comments about the dataset

Table 5 ChargingSession table

Field	Description
id	The unique identifier of the charging session
plug_in_soc	State of charge at plug-in time
plug_in_datetime	Date and time when the vehicle is plugged in
plug_out_datetime	Date and time when the vehicle is unplugged
charge_end_datetime	Date and time when charging was completed
charge_end_datetime_presence	True if the actual charge end datetime is available, False if it is set equal to the plug-out datetime
energy_supplied	Total energy supplied during the session (kWh)

Table 6 ChargingStation table

Field	Description
id	The unique identifier of the charging station
orig_id	Original identifier from the source dataset
manufacturer	Manufacturer of the station
model	Model name of the station
type	Type or classification of the station
num_plugs	Number of connectors available
max_charging_power	Maximum charging power (kW)
max_discharging_power	Maximum discharging power (kW)
dataset_id	Link to the corresponding dataset entry

Table 7 User table

Field	Description
id	The unique identifier of the user
orig_id	Original identifier from the source dataset
ev_id	Unique identifier of the electric vehicle
ev_manufacturer	Manufacturer of the EV
ev_model	Model name of the EV
ev_battery_capacity	Total battery capacity (kWh)
ev_battery_type	Type of battery used
ev_battery_useable_capacity	Usable capacity of the battery (kWh)
ev_v2g	Indicates V2G support
ev_max_charging_power	Max charging power the EV can accept (kW)
ev_max_discharging_power	Max discharging power the EV can provide (kW)
dataset_id	Link to the corresponding dataset entry

Table 8 ChargingStationForecast table

Field	Description
id	Unique identifier of the forecast
date	Date of the forecast
energy	Predicted energy use on the forecast date
connections	Predicted number of connections
experiment_id	ID of the MLflow experiment
run_id	ID of the MLflow run
created_at	Date when the forecast was generated

Table 9 UserForecast table

Field	Description
id	Unique identifier of the forecast
date	Date of the forecast
energy	Predicted energy use on the forecast date
duration	Predicted duration of the charging session
experiment_id	ID of the MLflow experiment
run_id	ID of the MLflow run
created_at	Date when the forecast was generated

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Data availability
Source code of Ev-Insights is available as open source at https://github.com/EV-Insights Datasets used in the manuscript are publicly available at: dataset_name, dataset_url ACN_Caltech, https://ev.caltech.edu/dataset AMB_Barcellona, https://zenodo.org/records/5721233#.YtaAc4TP0uU Elaad, https://platform.elaad.io/download-data/ OLEV, https://www.data.gov.uk/dataset/5438d88d-695b-4381-a5f2-6ea03bf3dcf0/electric-chargepoint-analysis-2017-domestics BeLib, https://www.data.gouv.fr/fr/datasets/belib-prises-de-recharge-pour-vehicules-electriques-disponibilite-temps-reel/ Harvard_dataverse, https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/NFPQLW Norway_12loc, https://zenodo.org/records/12730566

Declarations

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